

DHRUV KANDPAL

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PROFESSIONAL SUMMARY

Analytically minded Computer Engineering student (NMIMS, expected 2027) with hands-on experience in statistical modelling, data pipeline design, and analytics-driven application development using Python, SQL, and Java. Designed a 7-parameter statistical model for resource prioritisation under deadline constraints, validated across 1,000+ simulations and submitted to an IEEE-track journal (under review). Comfortable turning raw, messy data into structured, quarriable insight from relational schema design to building a retrieval pipeline over unstructured documents end-to-end.

TECHNICAL SKILLS

- Data & Analytics: SQL, Google BigQuery, Python (Pandas, NumPy), Scikit-learn, Statistical Modeling, Data Pipeline Design
- Languages: Python, Java, JavaScript, SQL, C++
- Databases: MySQL, SQLite, Firebase Realtime DB / Firestore, Google BigQuery
- Backend & Systems: Node.js, Socket.IO, Redis, RESTful APIs, RBAC
- AI/ML Tooling: Ollama, ChromaDB, Sentence-Transformers, Scikit-learn

PROJECTS

Bajao Bhai — Real-Time Collaborative Music Queue

2025

Node.js · Socket.IO · Redis · SQLite

- Engineered a real-time data-synchronisation pipeline using Socket.IO pub/sub over Redis to keep playback-state data consistent across every concurrent client in a shared session.
- Layered synchronous SQLite transactions under the async event pipeline to guarantee atomic data integrity on concurrent writes, eliminating race conditions when multiple clients updated shared queue data simultaneously.

RFM Customer Segmentation Analysis

2026

SQL (Google BigQuery) · Power BI

- Built a SQL pipeline in Google BigQuery consolidating 12 months of sales data and computing Recency, Frequency, and Monetary metrics per customer using NTILE window functions for decile scoring.
- Segmented customers into 8 actionable tiers (Champions, At Risk, Lost/Inactive, etc.) from the combined RFM score and visualized the segmentation in a Power BI dashboard to guide targeted marketing decisions.

Insurance Document Q&A — Local RAG System

2026

Python · Streamlit · ChromaDB · Ollama (Mistral)

- Built an end-to-end data pipeline for unstructured document analytics — custom chunking, embedding, and retrieval logic — deliberately avoiding LangChain to demonstrate pipeline-level understanding over framework fluency.
- Structured and indexed insurance policy PDFs into a queryable vector dataset (ChromaDB) and served a local Mistral model (Ollama) to deliver natural-language Q&A over that data end-to-end.

RESEARCH

UAAPS Framework — Urgency-Aware Adaptive Pathfinding Algorithm

IEEE Journal of Communications & Networks (Under Review)

- Designed a 7-parameter statistical model for resource-prioritisation data, applying convex penalty structures and binary threshold modelling to quantify urgency under deadline constraints.
- Ran 1,000+ simulations to statistically validate the model's outperformance over linear baselines — directly applicable to data-driven task-scheduling and SLA analytics.

LEADERSHIP & EXPERIENCE

Co-Head, PR Department — Sports Committee

NMIMS • Jun 2024 – Feb 2025

- Led a team of 20; drove a 5× increase in page views through a data-backed content strategy across official social handles.

Executive Member — Microsoft Learn Student Ambassador (MLSA)

NMIMS • Jun 2023 – Feb 2024

- Managed end-to-end design workflows (Figma, Canva) for multi-channel event promotions; maintained consistent visual brand identity.

EDUCATION & CERTIFICATIONS

B.Tech Computer Engineering — NMIMS, Navi Mumbai

Expected 2027 • CGPA: 2.8/4.0 (Sem VI)

- Google Data Analytics Professional Certificate — Coursera (In Progress)
- Google Cloud Study Jam Program (Jul 2023)